

# Active Inference BookStream #002.1 ~ Thomas Parr

## ~ Active Inference and Free Energy Principle...

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hello and welcome it's February 20th 7th 2023 we're here in active inference bookstream number 2.1 with Thomas Parr as well as Terry and Ali so thank you all for joining this should be a very fun discussion and let's go first by way of introduction and beginning to Thomas welcome and please how did you come to write this textbook in collaboration with pizulo and friston well again thank you very much for having me here um how did we come to write the textbooks good question I think um I think Giovanni was the the person who really had the idea and pushed for this that I think um Carl has already previously been approached by um by the editor who ended up managing application of the book um and when Giovanni had the idea that this would be a really useful contribution um he and I had a number of discussions about it both in in London and in Rome where he's based um and sort of put out a plan and and sort of came up with something that we hoped would be useful for the community I suppose part of the part of the reason why we thought it'd be a useful thing to do is that active inference is a very multi-disciplinary type field that draws a lot from lots of different domains and I think one of the one of the struggles you know we all had when we started out in it and I think certainly what the students and uh people who are interested in the topic say to us is that it's often quite difficult to trying to find all of these different topics trying to find the right background when they don't all come from one field so putting it all together in one book with a consistent notation consistent language and trying to make it complete we hoped would just make the field a little bit more accessible um to others who wanted to get involved how did this dovetail or integrate with your own Education and Research directions like what phases or stages were you moving through well this discussion was happening well I think so when we initially started discussing it it must have been probably the last year of my PhD and so by that stage I'd spent uh three or four sorry two to three years um sort of getting to grips with active inference um having spent you know being relatively recent in terms of learning some of the key material myself I think it was quite a good stage to be at in terms of thinking about what I

would have wanted to know a few years previously um the actual writing of the book then carried on over the next year or two um and so by that stage I was I was back in clinical training as well which I think is also a useful perspective in terms of some of the later chapters and thinking about application of active infant site models in the context of clinical conditions and Psychiatry and neurology and those sorts of areas that's actually a really interesting note how did the clinical experiences give a perspective on the book or how did the book or you're learning in these abstract and theoretical domains give a different Insight in the clinical setting it's a good question and I think you're right to highlight the bi-directionality of it that both the useful in in different ways um I think for me the clinical setting is quite useful in terms of thinking you know first of all in interacting the different sort of different sort of um audience so people who may be much more skilled on the clinical side of things maybe add less experience on the mathematical computational side of things but may still find the concepts very useful so from that side of things it's quite a useful exercise in being able to communicate these things um and then there's sort of the interaction on the patient side of things and trying to you know I think it makes a big difference first of all in interacting with patients has an impact on how you might think about the interesting questions and the things you might want to use them to entrance for but also having an understanding of principles behind it helps you I think formulate your questions about how individual patients might present or the reasons for that I think is a useful way of of thinking about I mean I suppose on two levels so one one in terms of understanding understanding particular sorts of pathology in terms of the inferences that might have gone wrong um but also in terms of understanding your own active influential processes when you're trying to work out what's going on yourself asking the right sort of questions to gather the right sort of information and and test those internal hypotheses I think that's a very useful way of thinking about being a doctor awesome yeah I like to say about that the Bayesian optimal differential diagnosis Ali or Terry on any of these thoughts do you have any questions or comments you want to raise there or any follow-ups uh well yeah actually I just wanted to ask uh what are the intended audience of the book and what do you think are the minimum prerequisites for going through the whole book because I'm sure you'll agree that even today the intersection between the people who are well versed in mathematics and the people who have a solid background in say neurobiology are quite narrow so so what's the solution here and what um did you think about about the organizational structure of the book that could accommodate various uh audiences so um yeah again another really good question and I think you're right to highlight that that intersection is relatively narrow and there are lots

of lots of people who I think I struggle to feel involved in in those kinds of discussions um and you're absolutely right to say that and I think you know what one of the solutions is is the sorts of things that you guys are doing in terms of setting up these reading groups trying to open that up trying to provide educational resources I think that's really very important um in terms of the specific questions about the book itself who's the audience well and you know what are the prerequisites I think it sort of depends what you want to get out of it and what your purpose is in reading it um so some aspects of the book I suppose will appeal more to some audiences and others to to other kinds of audience um I would hope that the readership or the audience is relatively broad because we've tried to provide as much background as we can on sort of the mathematical and the neurobiological side of things to try and complement the skill set of whoever's not in that intersection who comes to it and who who may be need a bit more sort of neurobiological grounding or who may need a bit more mathematical and Technical grounding um but I think ultimately you like with most textbooks you don't necessarily need to come to it thinking I'm going to understand everything in this book when I come away from it some people may want to um but other people depending upon what they want to get out of it they might say actually what I really want to get out of this is an understanding of some of the the kind of message passing as it's manifests in the brain whereas other people might come to it and say actually I already understand a bit about that and what I want to get out of this is to understand a bit more of the technical detail um you know if you're a robotic is coming to this it probably doesn't matter too much about how things are implemented in the brain if you're a neurobiologist coming to this you may not need to know the details of the implementation as long as you can understand it conceptually and I hope the organization particularly putting things like dependencies into detailed information at the back tries to keep most of the text a bit more accessible the early chapters being relatively light on the technical material and building up the concept Concepts before then moving into the more um technical aspects about the uh chapters let's actually just take a quick look at the chapters um we're in the uh Coda document that we use so perhaps you could just say one or a few sentences on each chapter what's the short of each chapter what is it doing in this 10 chapter work you're testing my memory now the titles are there the cues are there titles epistemic cues are there okay um well the overview is what it says on the team it's a description of what's to come in the subsequent chapters to try and try and make orientate people and the low range and high rate I think again Giovanni has to take credit for this idea of saying actually there's quite a it's often difficult to know where to begin when you're trying to understand um active interference

in the free energy principle and the Bayesian brain and all those sorts of things because it seems like there are so many potential starting points all of which ultimately I think get at the same thing but knowing where you begin is is often really quite difficult um and so these two chapters were designed to explicitly acknowledge that um that difficulty um so the idea of the low road is to say well let's take a pragmatic perspective um where we take ideas that have developed in in Neuroscience based upon our understanding of of the brain and of psychology and models that have been developed to explain those things and let's take those to their conclusion which is in a sense the free energy principle or which is active inference the idea of um using our internal models both to drive behavior and to understand what's going on in the world around us um the high road perspective was designed to take a different approach to that and instead of starting from what's already been developed and the sort of natural trajectory in the neurosciences um the high road chapter was more saying let's start from those principles as much as we can um and here was a tricky balance to try and make sure that we we didn't go too far in depth that we'd put people off by the by the third chapter but enough that people might get a bit of a conceptual sense of where you might go with this uh and and and here we we sort of Started from the kind of physics of sales organization the ideas that come underneath that and how those ideas can then feed back into this same idea of um of accidents that the neurosciences in Psychology seem to have arrived at relatively naturally suppose chapter four is when things start getting a bit more technical um and starts dealing with the specifics that which I think most people would want to know if they were implementing these models themselves um so trying to understand exactly how you construct an internal model how you specify how to write it down mathematically um how you how you perform inference with those how you understand the message passing schemes chapter 5 then then takes a step back and thinks well in the context of the brain in the context of neurobiology what do those things actually look like how do we understand those inferential processes with different kinds of generative models in terms of what we know about message passing and synaptic communication and and that those those five chapters together really are the the first Partners book and the the subsequent five chapters really then take you through the idea of actually applying those ideas in particular domains um so chapter six is sort of the overview chapter looking at the rest of it and it's really setting out if you wanted to sit down and design a model either conceptually your competition these are the steps you would have to go through to construct that and then chapters uh seven and eight take you through that sort of recipe it for several examples building up different concepts and both the models designed in discrete time

where you're formulating things as a sequence of events and models formulated in continuous time we we work with continuous dynamical systems and differential equations chapter 9 as I mentioned before you know one of my interests is in the clinical domain and trying to understand what sort of models people use to perform inference how those can go wrong and part of part of the research on that is thinking well how do we then fit these models and use them to actually draw inferences about what's going on particularly individuals or groups of individuals and then chapter 10 takes the broadest perspective and and looks at how that relates to a range another series and and ideas both in the neurosciences and Beyond and then I'm sure dependencies speak for themselves awesome great overview let's come back to our questions so you laid out that the first half of the book is about learning and the second half of the book is about applying and it begins with that recipe so if chapter 6 is the recipe as a chef and sous chef yourself what else have you found was important in the restaurant what is your full stock active inference modeling look like and maybe summarizing in six or going beyond what was in six like what have you found to be important as part of the modeling Journey especially with teams where people might have like different levels of domain specific knowledge or active knowledge I like the restaurant analogy actually and I was thinking I suppose the first thing that anybody needs to do if they're setting up a restaurant or a model is thinking about who their customers are and what they might want and so in the scientific domain you're thinking well what what sort of Journal am I planning on submitting to this to what sort of questions are the communities who read those Journal articles interested in and I think that's a really important First Step so that you know what I'm actually trying to do with this model what's not such a good idea often although you can find interesting things by doing it is to say well I'm going to develop a model and work out what it's for later on you can end up sort of tying yourself up a little bit by doing that although there is value in exploring us I think we would all agree and then I suppose in terms of ingredients It's relatively straightforward and deliberate deliberately so I suppose you know the idea of of these sorts of models is that they should be relatively accessible relatively easy to construct with um without having to make use of large supercomputers or or anything like that and a lot of the models that I develop can be run on a laptop fairly straightforwardly um so you know obviously just picking your Computing language and how you're going to construct that there are now a couple of different options out there um Matlab being the one that I think is most well developed but now lots of people um developing python-based implementations of accidents and schemes and particularly in the Markov decision process

Realm um so that gives you I suppose that some of the key ingredients but then it's really sitting down and thinking well how how would I now actually construct this model and the key question I think often is if you have a particular phenomena or a particular task in mind that you want model to perform it's thinking well how would I go about actually constructing the data that's presented to a participant performing that task all presented to an agent and that's trying to perform this task and if you can explain how those data are generated then you effectively have the generative model that that implicitly your agent or your participant or whoever else should be using to be able to perform that task so really all you need to be able to do is to work out how to create the experimental stimuli and once you've done that you've almost already created your model and the rest comes from predefined pre-existing inversion schemes that actually minimize the free energy for that model um so at that stage it becomes relatively straightforward I say relatively because as anybody who's has experience encoding knows that a lot of time is spent debugging and throwing our barriers and spending a while trying to work out why with a bit of Googling to work out a solution around that um and I think those are all important parts of the process awesome Ollie or Terry on that recipe and modeling question any thoughts on um yeah uh actually uh I also wanted to ask if um uh do you have a plan for uh extending the uh codes or supplementary materials to include other languages such as imdp packages or um any other packages that might come up later and would be widely used in active inference because as far as I know uh there isn't any specific GitHub repo or something dedicated for this book so far so do you have any plans in developing such a supplementary material for the book we don't at the moment um it might be an interesting thing I mean it might be a useful thing to for to be done by by the actual Insurance Community as well as a way of sort of being engaged involved in it I suppose one of the difficulties of doing that for a book is that often these things move on so quickly by the time it comes out in a book that those packages have moved on um so I think you know it was it was useful for us to give a simple example of Matlab implementation to demonstrate the principles in in one of the packages that we've been involved in developing as part of the SPM package where we've been involved in developing the active routines as part of that I think it you know it's it's certainly a good idea thinking about having other um uh repositories out there with all sorts of demonstrations so the sbm package does have a range of different demonstrations for different sorts of um uh implementations as well for different kinds of models which is a really helpful resource and I I often suggest to people one of the first ways of getting a sense of how to build these models is to look at one that's already been built and

and do their best to break it change things and make it behave in um and so I think you're absolutely right to say examples of the code and how it's implemented are vital in bringing people into this um but yes I think you're absolutely right to say that it'd be really helpful to have both you know Python and Julia implementations and all sorts of different ways of doing it and examples so that people can pick those up I agree completely and of course one of the advantages of python and Julia here is is that they're open source and you don't necessarily need academic license to use it which again I think increases the number of people who can access it giving an overview to someone who might not have been following in the blow by blow of our little literature Corner over the last few years how would you say active inference modeling has evolved during your time in the space where are we at just generally speaking with applications what applications are developed or possible today that when you were starting your PhD seemed like they were just implausible so I think if we go back a bit I suppose some of the earlier implementations were almost all in the continuous domain um so they're all based upon um things like the generalized filtering type algorithms that were initially developed as sort of filtering out um methods to perform time series um those things then equipped with an active component then then the canes and the earlier active inference type simulations and a lot of that focused on things in the movement domain whether that be eye movements or um or control of Limbs to the extent that you can sort of develop handwriting and those sorts of things one of the interesting things that started to happen based upon that was to think about how how you then get something that looks more sequential emerged from those continuous Dynamics and handwriting was actually a really good example of that where there were some simulations developed in in which in which the continuous model um effectively had something a bit like a logical Terror dynamic in it and the idea bit there being that you can sort of um synological Terror Dynamics are a kind of dynamical system often used for modeling things like Predator prey interactions and the idea is that you get a peak in terms of a predator population sorry prey population followed by peak in the pre the Predators as they eat the prey causing drop in the prey population but then as the Predators die out of the prey come back you get this sort of sequence of Peaks from Predator prey Predator prey and you can generalize that to then have you know multiple different populations each preying on one another so you get this sequence of Peaks over time and that sort of emergence of sequences from continuous Dynamics I think was one of the key moves early on in terms of thinking about things that are a bit more cognitive in terms of how do I do this and then this and this and start to plan and generate much more purposeful type of

behavior so that that went so far until um until things moved much more to an explicitly discrete or sequential style of modeling that's when things like Markov decision processes started to be employed in the active entrance domain or partially observed Markov decision processes so those are models that say I assume my world is a sequence of States over time uh one after another Each of which generates some observations and from those observations I can make inferences about the states you can then say well if I can act upon the world I can change the sequence of States um as they evolve over time and by selecting different actions I can or different ways I can change that sequence and effectively planning and making decisions so that's again where you get into a much more cognitive sort of domain um some of the key advances um sort of early on while I was doing my PhD with the emergence of things like deep hierarchical models of things like Markov decision processes so here the idea was that in the world around us there are lots of things that evolve over different sorts of time scales and so one of the early examples of this was reading the idea that when you move your eyes around the page you're sort of looking at an individual word or the letters within that word that that word is itself part of the sentence and the time it takes you to infer what the word is is going to be much shorter than the time it takes you to work out what's going on in terms of the sentence the sentence and the paragraph and the paragraph on the page and the page in the book Etc and so there's a whole hierarchy of different time scales in the world around us and one way of accounting for that relatively simply in these sorts of models is to Simply stack them on top of one another so the outcomes you're now predicting from a from one of the mark of decision process type models I was describing um now reflects a sequence of very fast States and each of those States yes exactly is the the graphic of this and then each of those um outcomes predicts another sequence of very fast States so what you're left with is at the very top level a very slow progression of state after state after state for each of those states which might be the sentence you then have the word the words in that sentence beneath it and then for each of those words you might have each of the letters in that sentence oh sorry in each of the letters in that word and so that then sort of expanded the range of things that we can do in terms of how we give some sort of deep temporal structure into the kinds of models we develop sometimes people refer to that as kind of breaking the Markov property so for those who aren't familiar with this the Markov property is the idea that um that each step in time depends only upon the step immediately previous to it that there's no sort of that all you need to know about the next time step is what's happening in the current time step nothing else in the past is useful once you know that um



and people often suggest that means that these processes are effectively memoryless because it's only where I am now that matters is not what was happening well into the past by having a hierarchical structure where you have a higher level stage that is more slowly changing that's contextualizing what's going on at the lower level it means that the fast sequence of changes is actually dependent upon something much slower that's happening and that in itself is dependent upon something much slower and that gives you this a sort of memory it allows for the Markov property to be broken foreign so like I said we're sort of starting off with continuous the emergence of discrete from continuous the explicit modeling of continuous models decision making and the sort of exploration and exploitation type trade-offs that you get once you you treat planning as being an influence process the idea that you can then stack these things hierarchically and have a range of different time scales that break the Markov process allow us to think about things like memory and working memory and I suppose one of the next stages was the combination or the reintroduction of the continuous models as the lowest level of one of these hierarchies the idea that actually it's all very well being able to make decisions but at some stage you need to implement those decisions in a world that's made up of physics and muscle lengths and um continuous sensory inputs and so putting these sort of Hybrid models together where you can use your decisions to then predict short sequences and trajectories defined in a continuous domain and I suppose since then they've you know been a range of further developments in action inference in that um and a range of developments in in a number of other directions um sort of in parallel so some of those developments have been thinking about implementation in other languages some of those developments have been thinking about combination with methods from Deep learning and instructional variation a lot of encoders and function approximators and amortized inference um which a lot of people in the robotics Community have been pioneering particularly so that would be one One Direction that things have been developed in and and I suppose the other direction that's seen a lot of development recently it has been the sort of underlying principles the um the physics that underwrites act again for instance the principles and how how you justify the sort of Dynamics and where where free energy comes from as a as a concept and why it's useful in understanding how um systems behave so that I hope that wasn't too much all at once but there was a sort of run through of some of the developments that um that spring to mind over the last few years awesome Terry or Ali any comments on that quick historical review oh again yeah I had another question regarding um actually it's a question curiosity uh did you have any specific reason for um adopting this particular mathematical notation

throughout the book because it seems like it somewhat diverges from the notations used in some of the well-known papers uh in the literature and uh but I know uh active inference notation uh is not consistently used across the literature but even in the papers that came out after the publication of this book written by Professor person or others um I don't see this specific notation used a lot in those papers so I just wanted to ask about the reasons behind this choice I think you're absolutely right to say that it's not always been used consistently um and it's worth saying that in some there are a couple of different tensions that that affect this so one is the idea that in a lot of papers that are published um because it's only dealing with an aspect of active influence and not trying to to try to put it all together they can sort of use a notation that may be the same as notation you might use in another field um without worry that people will will get conflicted and misinterpret one variable for another because everything's just defined within that one domain and another aspect to it is that in depending upon the paper in question and the audience is aimed at sometimes the notation is designed to um to match what what might be used in that broader field um whereas we've tried to try to make it as consistent as possible across different fields were there any particular notations that you were thinking help clarify uh yeah so for example um as just a very simple example using  $\eta$  for external States instead of  $X$  that's been used or  $S$  and  $a$  for um active States and sensory States or other kinds of uses of lowercase matrices and uppercase matrices which is used differently in a paper such as a step-by-step tutorial on active inference and these kinds of notational divergences okay um so yes I mean there's some quite nice illustrations so things like  $a$  and  $S$  which as you say are often used in some of the more physics orientated papers unfortunately if you use those in a mock-off decision process paper people will interpret those as being States and parameters of a likelihood Matrix so unfortunately we couldn't use those same letters without causing more confusion so what we try to do there is to to focus more on the um sort of practical inference side of things of you know the actual implementations we tend to use of active elements in particular problem settings and there when you're constructing a generative model um particularly if it's in the continuous domain we'll often use  $x$  as being the states that I have to infer which from the perspective of the physics type papers are the external States the things that are having an effect on my sensory States um we we've I can't remember whether we used  $y$  or or or  $o$  for the sensory States or the data in the book and it's why as you think is why okay um and that that sort of fits with the idea of using it as in the continuous domain where in um in a lot of settings when you're trying to predict um sensory data often why will end up being the variable at the end of Newton um so I I appreciate that it can be confusing

sort of jumping between different resources um but what we stroke for in this one was internal consistency as much as we could and trying not to overload specific variables or symbols

thank you thanks bigger 4.3 we bound ourselves coming back to again and again it four sages chapters seven and eight and it lays out discrete time on top and continuous time on the bottom in terms of their structural similarities but also they have some really important differences so could you just as you see it how do the discrete time and the continuous time models have similarities and differences such that you can lay them out in a graphically or visually similar way yet there's also some key differences for example in what each of the nodes means so how are they similar and different and what did you hope to illustrate by laying out figure 4.3 this way so I suppose the reason that they can be seen as similar is that effectively what you're trying to do in both cases is to represent a trajectory we're trying to represent how something evolves in time and that's something maybe the state's  $s$  in the discrete time or they may be the trajectory of some in States  $X$  in the and when you're trying to say how do I actually put together a set of numbers that tell me how something's evolved or evolving there are a couple of different ways you can do that one is to say I'm going to tell you where it is at time one and then I'm going to tell you where it is at time two and then at time three etc etc and that's a perfectly legitimate and perfectly reasonable way of writing down a trajectory it obviously misses out all the times between time one and time two which may or may not be significant depending upon what the problem is you're dealing with uh the second way that we approached is to say well you can use what what's known as a Taylor series approximation or a Taylor series or a polynomial expression or that trajectory um so essentially what we're saying there is that you just construct a function that is that that follows that the the the shape of that trajectory over time and you can construct that function by saying well let's look at where I am now at this point in time and my position how am I changing at the time if you just took those two variables together you just have a straight line telling you where you are and where you're going to be if you then take account of your acceleration you may even now have a curved line which says if I'm here now I'm going to be there then um and as you add in more and more elements to this so the rate of change of the velocity and the rate of change of the rate of change of the Velocity Etc and just add in more and more rates of change you effectively get a more and more precise function telling you at least locally around where you are now where you're going to be at various other times um so that that difference the either using Taylor series coefficients which are just um successive rates of change or by using a sequence of States over time are what's Illustrated in in the lower

and upper um Graphics here respectively um each of those things has there as you said that they have a lot of in common in terms of their sculpture in terms of the num or the number of numbers you need to be able to describe them um but it has their advantages and disadvantages depending upon what you're trying to actually model and I think for me I tend to think of the the sequence of categorical States over time as being quite useful in thinking about anything to do with decision making deciding between Alternatives um thinking about sequences like language and words and or even sequences in terms of the eye movements you make looking in one place then the next and the next rather than continuously moving your eyes around space in a smooth fashion uh whereas the process of interpreting sensory data that's coming in from your retina or from your your stretch receptors or whatever else or in terms of actually driving action that changes continuous variables in the world there I think is much more useful to be able to describe things in terms of a continuous number and those sorts of continuous trajectories that you can get through using these Taylor series type approximations which often referred to in this context as generalized coordinates of motion cool well another figure that we come back to again and again tucked away in chapter nine but giving a lot of context is uh figure 9.1 on what is termed meta Bayesian inference so could you describe what is happening in this figure and is this exactly as every empirical ethologist is going to agree with is this controversial is this novel so what is it doing for octave inference when we conceptualize our empirical task structurally like this and do you think that that is exactly how ethologists have long seen this topic or is this something contentious I think so I'll I'll talk through the figure but then but then you can tell me if that's how ethologists traditionally look at it and whether whether it's contentious I'll be interested to so the the idea behind this figure is to to highlight that when when dealing in psychology or Neuroscience or Fields where you're trying to understand our particular system is behaving or how it's essentially what's going on in the mind or the brain or the nervous system of of some creature that is behaving in some way um there are two levels of which you can look at it one is saying well from my perspective as the experimenter what are the data I'm measuring and that may be behavioral measurements from that creature it may be taking electrophysiological measurements it may be asking it a question if it's a patient in a clinical setting I'm essentially now trying to optimize my model of this system now in the neurosciences and in the settings I've outlined above the the um the kind of system we're interested in is the sort of system that has its own model of other sorts of system and so the kinds of questions we're interested in the kind of parameters we're interested in are things like the prior belief that this creature has

about this thing or how precise it thinks it's data are and so the parameters I'm now asking a question about are the parameters of the beliefs of another of another system so the outside box here um the the one that the parameters are going into the experimental stimuli are going into and The observed Behavior coming out of um is essentially my model of how this other system behaves now the inside box there the one in the dotted line or the dashed line um that now represents the model that that system has to understand the world around it um and so when when we construct our model that contains another model inside it um and by saying if this creature believed this this is how they would make inferences about their environment and generate Behavior I can now predict the behavior I would observe if this creature believed this thing um so the subjective objective idea is that the objective model is is the experimenter um and its beliefs and the experimenters beliefs and the observations they make by measuring what the participants or experimental subject does and the subjective model is the participants model of the world around them so that I think would be my summary of how this figure works and the the idea it's trying to get across now you can tell me how ethologists May disagree with that or agree with it or what they'd find controversial I'll give a thought and Ali or Terry feel free to give a follow-up or a thought I don't think in principle it's contentious or dissentious however there might be an explicit or an implicit belief course ironically that we can simply take some of these parameters coming into our experimental setup and even the selection of experimental stimuli we can take those as simply given which is a very frequentist perspective and then on the observed Behavior we can simply use descriptive statistics we'll characterize the predefined groups according to the one experiment we ran we'll do the t-test we'll do the Anova we'll come up with a p-value it's going to be frequentism all the way forward so even if we're doing frequentism about a generative model there's a sense that if we just clamp down on our experiment and treat it as handed down without error instead of being actually drawn from distributions of possible experiments we could have done if we clamp down Upstream of the experiments we'll just be able to describe and ultimately publish without needing to look at what else could have happened and I think this diagram and also the very subtle dashed line around even the bigger box suggests that just like the active clinician engaging with a conversation or the active experimenter also engaging iteratively over their system of Interest that itself is an Adaptive epistemic process where yes you could simplify it by treating the parameters and the stimuli sequences as simply given but then you don't have access to the generative model that actually is the generative process for the subjective model so summarize in principle uncontentious however sometimes discordant with

the ways that behavioral experiments are conceptualized planned and analyzed one reason why it's helpful because during the book we we try to stay on the road on the high road of the low road but when we're in discussions it's almost like there's a whole road map and people have their own backgrounds and questions and it kind of comes together with all these different topics that people are curious about ultimately that bring them to active inference so going to this metabasian figure helps in some ways situate how the authors are seeing this entire process experimental design observation and cognitive modeling and that was unpacked also in the recent map territory fallacy fallacy work Ali or Terry any thoughts or questions in our last 10 minutes is there a figure you want to look at or we can go to some other places maybe we can ask the missing figure in uh it was in I think it was in figure seven point yes yes or something 7.2 in the lower left so then this was actually corrected by I I don't know quite how it ended up Vanishing but it was corrected in one of the later print versions I'll see if it's in the version of course in front of me now and I can show you what it would have looked like um but apologies for that uh which figure is it 7.2 on page 128. and um unfortunately I picked up a copy of the book that doesn't have it um so there have only been one and a half fractal dimensions of printing yes um yeah I think unfortunately I don't think that one's corrected yet in the online copy um essentially what would have appeared in that that figure or what did in some of the later printing versions is the errors that are driving the updates of the states in the plot above um now because the so the errors here are um are effectively the um are the free energy gradients so uh or the negative three energy gradients so essentially the as you as you go down the free energy gradient you're effectively minimizing your error so what that means in practices that the errors would effectively be the rates of change of the states and the plot above so if you take the states and the plot above and you imagine what the rates of change of each of those lines are that's what you would get in the plot below well on this same kind of Topic in chapter seven we're in the discrete time generative models and um a little bit less so here but in the latter models here in 7.13 and in what we just looked at in 7.2 where are these continuous interpolations coming from if it's a discrete time model that's a good point so I I suppose there's an important distinction between if there's there's the distinction between um the things that are being inferred which are happening over discrete steps in time and then there's the inference process itself um which deals in probabilities which are themselves continuous they can vary in this case anywhere between zero and one um and the time over which that inference happens so to give you an example if you imagine that you're you the way you're analyzing this figure at the moment is by making a series of category movements

and you're not you're not sort of continuously moving your eyes from one block to the next you're making a series of rapid um ballistic movements that um and take you from one to the next um and effectively every time your eyes stand still you have a new data point that you're now trying to analyze you can imagine that that between that or every time your eyes stays still everything else in your brain doesn't just stop um so once you've got that new data point your brain now has some time to simulate that data point to draw inferences about what is seeing there and those inferences happen through the continuous dynamics of what's happening in in your neurons and various networks across your brain um so you can think of the continuous changes that are happening for each state as as not being changes in the world that we're trying to infer not being changes in the data point I'm currently looking at the changes in my beliefs about it before I then sample the next data point interesting I have a few more figure based quick questions but Terry do you want to bring up anything or just give any remarks no worries if not well I I am very much just trying to understand these Concepts so I I don't make any pretense uh having any deep understanding but the the key thing that interests me is how you jump from this sense of uh how we update our predictions into the clinical realm and I I I'm interested in pain medicine and I see neuroplastic pain is very much a mistaken prediction based on often uh misinterpretation by medical professionals and the information that they give to patients and I'm really intrigued by the potential for uh using these Concepts to to give patients an architecture in which they can start to understand that they can that they have some agency and altering their future predictions and really change in their experience of their pain and and as you're a clinician I'd be really intrigued to get your sense of where this is going in that type of print clinical Realm I think that's a really nice example as well thinking about um thinking about attitudes was pain and particular kinds of pain particularly when there isn't a sort of obvious tissue damage that can be identified that's um that's causing that pain that still is obviously a very real sensation to the people who are are experiencing it um and I think it generalizes to a number of other other conditions um and probably there's an overlap with things like functional neurological disorders probably a number of medically unexplained symptoms and things where that there is a huge amount of overlap in these sorts of processes where you can't necessarily directly identify a big structural lesion of any sort but clearly the way the nervous system is working um something isn't quite isn't quite right um pain I think is an interesting example particularly because pain is never a real thing out there in the world is it I mean it's it's always an inference and so I think that's one of the first things to say to anybody you know regardless of whether he got pain because

you've lost a leg or um whatever else it there's no such thing as pain as an external state it is just your inference your explanation for the the um signals that are coming in from your body now one of the important Concepts that I think comes into all of these um sorts of syndromes and probably many others as well is the idea of precision the idea that um the data I predict sensory data coming in um I I can in addition to estimating what those data are going to look like I also have to estimate how confident I am in what they're going to look like um if I'm not very confident or I think I'm dealing in something that's very noisy then I can actually discount a lot of that whereas if I believe that it's all extremely reliable very clean data that's coming in then I want to put a lot of weight on that and another way of saying that is that there are some things I pay attention to and some things I don't you know just saying it's reliable data is just a matter of saying I'm going to attempt to it now it might be that in a number of these different syndromes um what's going on is that our bodies have um overestimated the Precision overestimated how clean the data are coming in from a particular body part um and that that's why we then over interpret any signals that are coming in as if they are painful and the influences we draw from those signals are just too confident and too driven by something that they shouldn't be an example I often like to use in in clinical discussions is that it I told you to imagine to pay attention now to the the feeling that your left sock is having on your left big toe you can probably feel it um but you were completely unaware of it before but now if I say continue to pay attention to that and carry on paying attention to that and and you're unable to pay attention away from it anymore if you felt that all the time it would feel unusual it would feel wrong um maybe a misestimation of precision in that way maybe what's going on in a lot of these sorts of conditions partly I think it's often interesting that in in some of these people there's um there was at some stage an injury or something that affected that body part and you can imagine that if you went through a period of illness where when you did have tissue damage and we're in pain for that reason it might direct your attention to that and you may have a learned deployment of attention towards that area that might persist even after um even after the the insults and the damage was resolved um and again maybe maybe it's that sort of persistent uh mislearning of um how precise things are that leads to the continued experience of pain even when there's nothing um sort of externally causing it anymore so I think just having a sense of inference and of um and the idea that that Allah perceptions of the world around us are ultimately inferences um says that you know if we change our model in some way we're able to adapt and change and develop plasticity in our our brains and spinal cords can be exploited uh then then we can also um correct those mislearned models um and



hopefully resolve those sorts of symptoms and syndromes thank you well we are at the very close really appreciative Thomas of how much we were able to cover what are your closing words or thoughts potentially to someone who has listened to this one hour discussion as their appetizer before joining the par at all restaurant service well I know I've I've enjoyed the discussion thank you for having me here I that it sounds I mean I'll reiterate what I said earlier that I think one of the key things in in making this field more accessible to more people are these sort of um uh reading groups the ideas of trying to create these educational resources and to unpack and discuss these things um and you know I'm impressed by all the infrastructure you've put together and um and by everything you've managed to achieve by doing this um so I'd certainly encourage anyone who's interested to join up to your reading groups um and I hope they they get something out of it and out of the book thanks Ali well uh yes um actually since the beginning of the first cohort um this book has practically become a part of our life that we love and cherish and uh it's amazing to see how with each iteration of our reading group uh it gives rise to whole different set of questions and and discussions and uh actually I'm a firm believer in this famous quote I don't know by whom which says that the sign of a great book is not how many answers it provides but rather how many thought-provoking questions it engenders so based on that Criterion and judging by the variety of questions uh brought up during our live discussions as a kind of empirical evidence for that I think it's probably safe to say that this is truly a great book so congratulations on that and thank you very much great closing thoughts so Thomas thank you again you're welcome back anytime thanks Ali and Terry for all the great participation and until next time bye